TOPIC 3:

```
'Q1) Write a console application that asks for your forename And then your surname And
then displays both names on the same line.
Module T3Q1
    Sub Main()
        Dim forename, surname As String
        WriteLine("Enter your forename")
        forename = ReadLine()
        WriteLine("Enter your surname")
        surname = ReadLine()
        WriteLine(forename & " " & surname)
        ReadKey()
    End Sub
End Module
```

TOPIC 5:

```
'T5Q1)Write a program that will read in three integers And display the sum.
Module T5Q1
    Sub Main()
        Dim a, b, c As Integer
        WriteLine("Enter number1")
        a = ReadLine()
        WriteLine("Enter number2")
        b = ReadLine()
        WriteLine("Enter number3")
        c = ReadLine()
        WriteLine("Sum = " & a + b + c)
        ReadKey()
    End Sub
End Module

'T5Q2) Write a program that will read two integers And display the product.
Module T5Q2
    Sub Main()
        Dim a, b As Integer
        WriteLine("Enter number1")
        a = ReadLine()
        WriteLine("Enter number2")
        b = ReadLine()
        WriteLine("Product = " & a * b)
        ReadLine()
    End Sub
End Module

'T5Q3) Enter the length, width And depth of a rectangular swimming pool. Calculate the
volume of water reT5Quired to fill the pool And display this volume.
Module T5Q3
    Sub Main()
        Dim l, w, d As Integer
        WriteLine("Enter length of pool in cm")
```

```
        l = ReadLine()
        WriteLine("Enter width of pool in cm")
        w = ReadLine()
        WriteLine("Enter depth of pool in cm")
        d = ReadLine()
        WriteLine("Volume of water reT5Q to fill pool in cm3 = " & l * w * d)
        ReadKey()
    End Sub
End Module

'T5Q4) Write a program that will display random numbers between 1 And 6 until a six Is
generated.
Module T5Q4
    Sub Main()
        Dim r As Random = New Random()
        Dim randnum As Integer
        Do
            randnum = r.Next(1, 7) ' Upper bound is exclusive
            Console.WriteLine(randnum)
        Loop Until randnum = 6
        Console.ReadKey()
    End Sub
End Module

'T5Q5) Write a program that will display six random numbers between 5 And 10.
Module T5Q5

    Sub Main()
        Dim r As Random = New Random()
        For i = 1 To 5
            Console.WriteLine(r.Next(5, 11)) ' Upper bound is exclusive
        Next
        Console.ReadKey()
    End Sub

End Module

'Pg24:
'T5Q6) Write a program that will ask the user for their first name. The program should
then concatenate the name with a message, such as 'Hello Fred. How are you?' and output
this string to the user.
Module T5Q6
    Sub Main()
        Dim name As String
        WriteLine("What is your name?")
        name = ReadLine()
        WriteLine("Hello " & name & ". How are you?")
        ReadKey()
    End Sub
End Module

'T5Q7) Write a program that asks the user to enter two double numbers And displays the
product of these two numbers to 2 decimal places, with user-friendly messages. (tip: see
page 19 For examples)
Module T5Q7
    Sub Main()
        Dim num1, num2, product As Double
        WriteLine("Enter num1")
```

```
        num1 = ReadLine()
        WriteLine("Enter num2")
        num2 = ReadLine()
        product = num1 * num2
        WriteLine(Format(product, "####.00"))
        ReadKey()
    End Sub
End Module

'T5Q8) Calculate the area of a rectangle where the length And breadth are entered by the
user.
Module T5Q8
    Sub Main()
        Dim l, w As Integer
        WriteLine("Enter length of rect in cm")
        l = ReadLine()
        WriteLine("Enter width of rect in cm")
        w = ReadLine()
        WriteLine("Area of rect in cm2 = " & l * w)
        ReadKey()
    End Sub
End Module

'T5Q9) Calculate the area And circumference of a circle, where the radius Is entered into
an edit box. The radius And area may Not always be integer values.
Module T5Q9
    Sub Main()
        Dim r As Integer
        WriteLine("Enter radius in cm")
        r = ReadLine()
        WriteLine("Area of circle in cm2 = " & Math.PI * (r ^ 2))
        WriteLine("Circumfrence of cricle in cm = " & 2 * Math.PI * r)
        ReadKey()
    End Sub
End Module

'T5Q10) Create an automatic percent calculator that a teacher could use after an exam.
The teacher will enter the total number of marks And the raw score. The form them
calculates the percentage (to two decimal places).
Module T5Q10
    Sub Main()
        Dim totalmarks, rawscore, percentage As Double
        WriteLine("Enter total number of marks")
        totalmarks = ReadLine()
        WriteLine("Enter raw score")
        rawscore = ReadLine()
        percentage = (rawscore / totalmarks) * 100
        WriteLine(Format(percentage, "###.00") & "%")
        ReadKey()
    End Sub
End Module

'T5Q11) Create a currency converter where the number of pounds Is entered And the
procedure calculates the number of Euros. Assume the exchange rate of £1 = €1.15.
'Adjust your program To allow any exchange rate.
Module T5Q11
    Sub Main()
        Dim rate, amount, result As Double
```

```
        WriteLine("Enter number of pounds")
        amount = ReadLine()
        WriteLine("Enter excahange rate. £1 = €x")
        rate = ReadLine()
        result = rate * amount
        WriteLine("£" & amount & " = " & "€" & result)
        ReadKey()
    End Sub
End Module

'T5Q12) Try to adapt program 2 to deal with a cylinder. Area of an open cylinder Is 2πrh.
Area of a closed cylinder Is 2πr(h + r). Volume of a cylinder Is πr2h
Module T5Q12
    Sub Main()
        Dim r, h As Integer
        WriteLine("Enter radius in cm")
        r = ReadLine()
        WriteLine("Enter height in cm")
        h = ReadLine()
        WriteLine("Area of open cylinder in cm2 = " & 2 * Math.PI * r * h)
        WriteLine("Area of closed cylinder in cm2 = " & 2 * Math.PI * r * (h + r))
        WriteLine("Volume of closed cylinder in cm3 = " & Math.PI * r ^ 2 * h)
        ReadKey()
    End Sub
End Module

'T5Q13) Convert a temperature entered in Celsius to Fahrenheit.
Module T5Q13
    Sub Main()
        Dim temp As Double
        WriteLine("Enter temprature in Celsius")
        temp = ReadLine()
        WriteLine("Tempratue in Fahrenheit = " & ((9 * temp) / 5) + 32)
        ReadKey()
    End Sub
End Module
```

TOPIC 6:

```
'Pg32:
'T6Q1) Write a program that will ask for a person's age and then display a message
whether they are old enough to drive or not.
Module T6Q1
    Sub Main()
        Dim age As Integer
        WriteLine("Enter your age")
        age = ReadLine()
        If age >= 18 Then
            WriteLine("You are eligible to drive")
        Else
            WriteLine("You are not eligible to drive")
        End If
        ReadKey()
    End Sub
End Module
```

```
'T6Q2) Write a program that asks the user to enter two numbers And then displays the largest of the two numbers.
Module T6Q2
    Sub Main()
        Dim largest, temp As Double
        WriteLine("Enter number 1")
        largest = ReadLine()
        WriteLine("Enter number 2")
        temp = ReadLine()
        If temp > largest Then
            largest = temp
        End If
        WriteLine("Largest number = " & largest)
        ReadKey()
    End Sub
End Module

'T6Q3) Write a program that checks whether a number input Is within the range 21 to 29 inclusive And then displays an appropriate massage. Extend the program so a message out of range will cause a message saying whether it Is above Or below the range.
Module T6Q3
    Sub Main()
        Dim input As Integer
        WriteLine("Input number")
        input = ReadLine()
        If input >= 21 And input <= 29 Then
            WriteLine("Number within range")
        ElseIf input < 21 Then
            WriteLine("Number below range")
        Else ' Input greater than 29 then
            WriteLine("Number above range")
        End If
        ReadKey()
    End Sub
End Module

'T6Q4) Write a program that asks for three numbers And then displays the largest of the three numbers entered.
Module T6Q4
    Sub Main()
        Dim largest, temp As Double
        WriteLine("Enter number 1")
        largest = ReadLine()
        WriteLine("Enter number 2")
        temp = ReadLine()
        If temp > largest Then
            largest = temp
        End If
        WriteLine("Enter number 3")
        temp = ReadLine()
        If temp > largest Then
            largest = temp
        End If
        WriteLine("Largest number = " & largest)
        ReadKey()
    End Sub
End Module
```

```
'Pg33:
'T6Q5) Write a program that lets the user enter a number between 1 And 12 And displays
the month name for that month number. The input 3 would therefore display March.
Module T6Q5
    Sub Main()
        Dim input As Integer
        Dim month As String = ""
        WriteLine("Enter month number")
        input = ReadLine()
        Select Case input
            Case 1
                month = "Jan"
            Case 2
                month = "Feb"
            Case 3
                month = "March"
            Case 4
                month = "April"
            Case 5
                month = "May"
            Case 6
                month = "June"
            Case 7
                month = "July"
            Case 8
                month = "Aug"
            Case 9
                month = "Sept"
            Case 10
                month = "Oct"
            Case 11
                month = "Nov"
            Case 12
                month = "Dec"
        End Select
        WriteLine(month)
        ReadLine()
    End Sub
End Module

'ALT:
Module T6Q5Alt
    Dim month() As String = New String() {"Jan", "Feb", "March", "April", "May", "June",
"July", "August", "Sept", "Oct", "Nov", "Dec"}
    Sub Main()
        Dim input As Integer
        WriteLine("Enter month number")
        input = ReadLine()
        WriteLine(month(input - 1)) ' "-1" becuz indexing starts from 0 not 1
        ReadKey()
    End Sub
End Module

'T6Q6) Write a program that reads in the temperature of water in a container (in
Centigrade) And displays a message stating whether the water Is frozen, boiling Or
neither.
Module T6Q6
    Sub Main()
```

```
        Dim temp As Double
        WriteLine("Enter temp of water in degree C")
        temp = ReadLine()
        If temp = 100 Then
            WriteLine("Water is boiling")
        ElseIf temp <= 0 Then
            WriteLine("Water is frozen")
        Else
            WriteLine("Neither boiling nor frozen")
        End If
        ReadKey()
    End Sub
End Module

'T6Q7) Write a program that asks the user for the number of hours worked this week And
their hourly rate of pay. The program Is to calculate the gross pay. If the number of
hours worked Is greater than 40, the extra hours are paid at 1.5 times the rate. The
program should display an error message if the number of hours worked Is Not in the range
0 to 60.
Module T6Q7
    Sub Main()
        Dim hrs, rate, total As Double
        WriteLine("Enter number of hours worked")
        hrs = ReadLine()
        If hrs >= 0 And hrs <= 60 Then ' hrs lie in give in range
            WriteLine("Enter pay rate")
            rate = ReadLine()
            If hrs > 40 Then
                total = (rate * 40) + ((hrs - 40) * (rate * 1.5)) ' total = (pay in 40
hrs) + (overtime hrs * overtime rate)
            Else
                total = rate * hrs
            End If
            WriteLine("Total pay = " & total)
        Else
            WriteLine("Working hours not in range")
        End If
        ReadKey()
    End Sub
End Module

'T6Q8) Write a program that accepts a date as three separate integers such as 12 5 03.
The program should display the date in the form 12th May 2003. Adapt your program to
interpret a date such as 12 5 95 as 12th May 1995. Your program should interpret the year
to be in the range 1931 to 2030.
Module T6Q8
    Dim month() As String = New String() {"Jan", "Feb", "March", "April", "May", "June",
"July", "August", "Sept", "Oct", "Nov", "Dec"}
    Sub Main()
        Dim d, m, y, preY As Integer
        Dim dsuffix As String
        WriteLine("Enter date")
        d = ReadLine()
        WriteLine("Enter month")
        m = ReadLine()
        WriteLine("Enter year")
        y = ReadLine()
        If y >= 0 And y <= 30 Then
```

```
            preY = 20
        Else
            preY = 19
        End If
        If d = 1 Or d = 21 Then
            dsuffix = "st"
        ElseIf d = 2 Or d = 22 Then
            dsuffix = "nd"
        ElseIf d = 3 Or d = 23 Then
            dsuffix = "rd"
        Else
            dsuffix = "th"
        End If
        WriteLine(d & dsuffix & " " & month(m - 1) & " " & preY & y)
        ReadKey()
    End Sub
End Module

'T6Q9) Write a program that accepts a date as three separate integers such as 12 5 03 And
then calculate Is this Is a valid date Or Not.
Module T6Q9
    Dim days() As Integer = New Integer() {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30,
31}
    Sub Main()
        Dim d, m, y As Integer
        Dim valid As Boolean = True
        WriteLine("Enter date")
        d = ReadLine()
        WriteLine("Enter month")
        m = ReadLine()
        WriteLine("Enter year")
        y = ReadLine()
        If m < 1 Or m > 12 Then ' If month is invalid
            valid = False
        Else ' month is valid
            If d < 1 Or d > days(m - 1) Then ' If day is less than 1 or day is greater
than the max days of that month
                valid = False
            Else ' day is valid
                If y < 1 Or y > 99 Then ' If year is out of range
                    valid = False
                End If
            End If
        End If
        If valid Then
            WriteLine("Date is Valid")
        Else
            WriteLine("Date is Invalid")
        End If
        ReadKey()
    End Sub
End Module
```

TOPIC 7:

```
'Pg35:
'Q1) Write a program that displays the word 'Hello' on the screen 4 times on the same
line using the for loop.
Module T7Q1
    Sub Main()
        Dim i As Integer
        For i = 1 To 4
            Write("Hello ")
        Next
        ReadKey()
    End Sub
End Module

'Q2) Write a program that prompts the user to enter a short message And the number of
times it Is to be displayed And then displays the message the required number of times.
Module T7Q2
    Sub Main()
        Dim input As String
        WriteLine("Please enter a message to display")
        input = ReadLine()
        Dim repeat As Integer
        WriteLine("Please enter the number of times the message should be displayed")
        'The value must not be negative
        repeat = ReadLine()
        For i = 1 To repeat
            WriteLine(input)
        Next
        ReadKey()
    End Sub
End Module

'Q3) Write a console application to calculate the 8 times table.
Module T7Q3
    Sub Main()
        Dim i As Integer
        For i = 1 To 10
            WriteLine(i & " times 8 is " & 8 * i)
        Next
        ReadKey()
    End Sub
End Module

'Q4) Write a console application that will ask for 10 numbers. The program will then
display the sum of these numbers And the average.
Module T7Q4
    Sub Main()
        Dim numbers(9) As Integer
        Dim sum As Integer = 0
        For i = 0 To 9
            WriteLine("Please enter number " & i + 1) ' "i + 1" becuz indexing starts
from 0
            numbers(i) = ReadLine() ' stores value in array
            sum = sum + numbers(i) ' sums the value
        Next
```

```
        WriteLine("Sum of 10 numbers = " & sum)
        WriteLine("Average of 10 numbers =" & sum / 10)
        ReadKey()
    End Sub
End Module

'Q5) Write a program that asks for a number, And displays the squares of all the integers
between 1 And this number inclusive.
Module T7Q5
    Sub Main()
        Dim input As Integer
        WriteLine("Please enter a value")
        input = ReadLine()
        For i = 1 To input
            WriteLine(i * i)
        Next
        ReadKey()
    End Sub
End Module

'Q6) Write a program to display the squares of all the integers from 1 to 12 in two
columns headed 'Number' and 'Square of Number'.
Module T7Q6
    Sub Main()
        WriteLine("Number   Square of Number")
        For i = 1 To 12
            If i < 10 Then
                WriteLine(i & "              " & i * i)
            Else
                WriteLine(i & "             " & i * i)
            End If
        Next
        ReadKey()
    End Sub
End Module

'Q7) Write a program that displays all the numbers between 1 And 10,000. How long did it
take to execute?
Module T7Q7
    Sub Main()
        Dim watch As Stopwatch = Stopwatch.StartNew() ' Initiates a new stopwatch
        For i = 1 To 10000
            WriteLine(i)
        Next
        watch.Stop()
        WriteLine("Time Elasped in Milliseconds: " & watch.Elapsed.TotalMilliseconds)
        ReadKey()
    End Sub
End Module

'Q8) Write a program that displays all even the numbers between 1 And 10,000. How long
did it take to execute?
Module T7Q8
    Sub Main()
        Dim watch As Stopwatch = Stopwatch.StartNew()
        For i = 2 To 10000 Step 2
            WriteLine(i)
        Next
```

```
        watch.Stop()
        WriteLine("Time Elasped in Milliseconds: " & watch.Elapsed.TotalMilliseconds)
        ReadKey()
    End Sub
End Module

'Pg36:
'Q9) Write a console application that will ask for 10 numbers. The program will then
display the max And min value entered (you will need to use an IF statement And variables
to hold the max And min values).
Module T7Q9
    Sub Main()
        Dim input, min, max As Integer
        For i = 0 To 9
            WriteLine("Enter an integer")
            input = ReadLine()
            If i = 0 Then
                max = input ' sets the 1st value as max
                min = input ' sets the 1st value as min
            End If
            If input > max Then ' If new value > max
                max = input
            ElseIf input < min Then ' If new value < max
                min = input
            End If
        Next
        WriteLine("Max Value: " & max)
        WriteLine("Min Value: " & min)
        ReadKey()
    End Sub
End Module

'Q10) n factorial, usually written n!, Is defined to be the product of all the integers
in the range 1 to n
'n! = 1 * 2 * 3 * 4 * …….*n
'Write a program that calculates n! For a given positive n.
Module T7Q10
    Sub Main()
        Dim input As Integer
        WriteLine("Enter an integer")
        input = ReadLine()
        Dim ans As Integer = input
        If input > 1 Then ' Note1
            Do
                input = input - 1
                ans = ans * (input)
            Loop Until input = 1
        End If
        WriteLine("Ans = " & ans)
        ReadKey()
    End Sub

    ' Note1: Purpose of this check is to assure that
    ' the num is not -ve. Becuz if it will be -ve then
    ' "input = input - 1" will decrease the num making
    ' it impossible to reach value "0" which is req to
    ' terminate the loop
End Module
```

```
'Q11) Write a program that asks for a number And converts this into a binary number. You
will need to use \ And Mod.
'Method1: Descending Powers Of Two And Subtraction
Module T7Q11
    Sub Main()
        Dim input As Integer
        WriteLine("Please enter a value")
        input = ReadLine()
        Dim power As Integer = 0
        ' Loops until highest power of 2 is selected
        While (2 ^ power) < input
            power = power + 1
        End While

        ' Declares an int array with length equal to power
        ' i.e: for 2^3 is the power of 2 in the 4th bit
        ' hence an array of length 4 starting from index 0 is
        ' declared by power = 3. so no need to write "power - 1"
        Dim ZerosOnes(power) As Integer

        For i = power To 0 Step -1 'Starting from the most sig bit
            If (2 ^ i) <= input Then
                ' power of 2 deducted from number
                input = input - (2 ^ i)
                ' bit turned on
                ZerosOnes(power - i) = 1
            Else
                ' bit turned off
                ZerosOnes(power - i) = 0
            End If
        Next

        ' Makes the bits into group of 4 bits
        Dim r As Integer
        ' gets remainder when no. of bits/4
        Math.DivRem(power + 1, 4, r)
        If r <> 0 Then
            For i = 0 To 3 - r
                Write("0")
            Next
        End If

        ' Loop to display array
        For i = 0 To power
            Write(ZerosOnes(i))
        Next

        ReadKey()
    End Sub
End Module

'Method2: Short Division by 2 with Remainder
Module T7Q11Alt
    Sub Main()
        Dim input As Integer
        WriteLine("Please enter a value")
        input = ReadLine()
```

```
        Dim r As Integer
        Dim binary As String = ""
        Do
            Math.DivRem(input, 2, r)
            input = input \ 2
            ' ^Int division is req becuz a decimal num i.e 0.5
            ' will be rounded off to 1 but the req is to remove
            ' the decimal part rather than rounding off
            binary = r & binary
            ' ^Reasone why "r & binary" is not "binary & r"
            ' Places the later remainders as the most sig bit
        Loop Until input = 0

        ' Makes the bits into group of 4 bits
        r = 0
        ' gets remainder when no. of bits/4
        Math.DivRem(binary.Length, 4, r)
        If r <> 0 Then
            For i = 0 To 3 - r
                Write("0")
            Next
        End If

        WriteLine(binary)
        ReadKey()
    End Sub
End Module

'Pg36:
'Q12) Write a program that reads in a series of numbers And adds them up until the user
enters zero. (This stopping value Is often called a rogue value.) You may assume that at
least
'1 number Is entered. Expand your program to display the average as well as the sum of
the numbers entered. Make sure you do Not count the rogue value as an entry.
Module T7Q12
    Sub Main()
        Dim input As Double
        Dim sum As Double = 0
        Dim values As Double = 0
        WriteLine("Please enter a value")
        input = ReadLine()
        While input <> 0
            sum = sum + input
            values = values + 1
            WriteLine("Please enter a value")
            input = ReadLine()
        End While
        WriteLine("Sum = " & sum)
        WriteLine("Average = " & sum / values)
        ReadKey()
    End Sub
End Module

'Q13) Write a program that asks the user for a number between 10 And 20 inclusive And
will validate the input. It should repeatedly ask the user for this number until the
input Is within the valid range.
'Make changes To your program so it will also work If the user does Not want To type In
any numbers And uses the rogue value straight away.
```

```
Module T7Q13
    Sub Main()
        Dim input As Double
        WriteLine("Please enter a number between 10 to 20 or enter 0")
        input = ReadLine()
        While input <> 0 And (input < 10 Or input > 20)
            WriteLine("Please enter a number between 10 to 20 or enter 0")
            input = ReadLine()
        End While
        WriteLine("Done")
        ReadKey()
    End Sub
End Module

'Q14) Write a program that displays a conversion table for pounds to kilograms, ranging
from 1 pound to 20 pounds [1 kg = 2.2 pounds].
Module T7Q14
    Sub Main()
        Dim i As Integer
        For i = 1 To 20
            WriteLine(i & " kg = " & i * 2.2 & " pounds")
        Next
        ReadKey()
    End Sub
End Module

'Q15) Write a program that asks the user to enter 8 integers And displays the largest
integer.
'Adapt your program so that the user can type In any number Of positive integers. Input
will terminate With the rogue value Of–1. Adapt your program so that it will also display
the smallest Integer. Adapt your program from If necessary, so that it works If the user
types In the rogue value As the first value
Module T7Q15
    Sub Main()
        Dim input As Integer = -2
        Dim max As Integer = 0
        Dim min As Integer = 0
        Dim values As Integer = 1
        Do
            WriteLine("Please enter a positive integer number.1")
            input = ReadLine()
        Loop Until input >= -1
        If input = -1 Then
            WriteLine("Program Terminanted.")
        Else
            max = input
            min = input
            While input <> -1 And values <> 8
                values = values + 1
                WriteLine("Please enter a positive integer number." & values)
                input = ReadLine()
                If input <> -1 Then
                    If input > max Then
                        max = input
                    End If
                    If input < min Then
                        min = input
                    End If
```

```
                End If
            End While
            WriteLine("Max value = " & max)
            WriteLine("Min value = " & min)
        End If
        ReadKey()
    End Sub
End Module

'Q16) Write a game in which the user guesses what random number between 1 And 1000 the
computer has 'thought of, until he or she has found the correct number. The computer
should tell the user whether each guess was too high, too low or spot on. (TIP: use the
Math library, which has a random function. See page 18)
Module T7Q16
    Sub Main()
        Dim input As Integer
        Dim rand As Random = New Random
        Dim r As Integer
        r = rand.Next(1, 1000)
        WriteLine("Guess the number b/w 1 to 1000")
        input = ReadLine()
        While input <> r
            Dim diff = r - input
            If diff > 100 Then
                WriteLine("Too Low")
            ElseIf diff < -100 Then
                WriteLine("Too High")
            ElseIf diff < 20 And diff > -20 Then
                WriteLine("Really Close")
            Else
                WriteLine("Close..")
            End If
            WriteLine("Retry")
            input = ReadLine()
        End While
        WriteLine("Spot On!")
        ReadKey()
    End Sub
End Module
```

TOPIC 8:

```
'Pg41:
'T8Q1) Write a program that reads 6 names into an array. The program must display the
names in the same order that they were entered And then in reverse order.
Module T8Q1
    Sub Main()
        Dim names(5) As String
        Dim i As Integer
        For i = 0 To 5
            Console.WriteLine("Enter name " & i + 1)
            names(i) = Console.ReadLine()
        Next
        For i = 0 To 11
            If i < 6 Then
                Console.WriteLine(names(i))
```

```
            Else
                Console.WriteLine(names(11 - i))
            End If
        Next
        ReadKey()
    End Sub
End Module

'T8Q2) Make a program that fills an array with 5 elements with values entered by the
user.
'Print out all elements In order.
Module T8Q2
    Sub Main()
        Dim array(4) As String
        Dim i As Integer
        For i = 0 To 4
            Console.WriteLine("Please enter a name")
            array(i) = Console.ReadLine()
        Next
        For i = 0 To 4
            Console.WriteLine(array(i))
        Next
        ReadKey()
    End Sub
End Module

'T8Q3) Make a program that first asks the user how many values to enter. The program will
then fill an array with all entered values. When all values are entered the program
prints out the highest entered value And its position (And index) in the array.
Module T8Q3
    Sub Main()
        Dim num As Integer
        Console.WriteLine("Specify how many values you wish to enter")
        num = Console.ReadLine()
        Dim array(num) As Integer
        Dim maxnum As Integer
        Dim index As Integer
        For i = 0 To num - 1
            Console.WriteLine("Enter value number " & i + 1)
            array(i) = Console.ReadLine()
            If i = 0 Then
                maxnum = array(i)
            End If
            If array(i) > maxnum Then
                maxnum = array(i)
                index = i
            End If
        Next
        Console.WriteLine("Index = " & index)
        Console.WriteLine("Max Value = " & maxnum)
        ReadKey()
    End Sub
End Module

'T8Q4) We want to simulate throwing a die 30 times And record the scores. If we did this
'manually' we would end up with a tally chart: If we use a computer to keep a count of
how many times each number was thrown, we could use an integer array (index range 1..6)
```

```
instead of the tally chart. In general, a die throw will give a score i, and we want to
increment the count in the i th element.
'TallyChart(i) ← TallyChart(i) + 1
'Write a program To simulate the throwing Of a die 30 times. The results Of the
Module T8Q4
    Sub Main()
        Dim tchart(5) As Integer
        Dim r As Random = New Random
        Dim i As Integer
        For i = 0 To 29
            Dim rnd As Integer = r.Next(1, 7) ' Upper bound is exclusive
            tchart(rnd - 1) = tchart(rnd - 1) + 1
        Next
        For i = 0 To 5
            Console.WriteLine("Number of " & i + 1 & "s = " & tchart(i))
        Next
        ReadKey()
    End Sub
End Module

'T8Q5) Write a program which asks the user for the subjects done in each period for each
day And then prints out the timetable with suitable headings.
Module T8Q5
    Sub Main()
        '5 Days and 8 periods
        Dim timetable(4, 7) As String
        Dim days() As String = New String() {"Monday", "Tuesday", "Wednesday",
"Thursday", "Friday"}
        Dim i As Integer
        For i = 0 To 4
            Dim a As Integer
            For a = 0 To 7
                Console.WriteLine("Enter subject taken on " & days(i) & " period " & a +
1)
                timetable(i, a) = Console.ReadLine()
            Next
        Next
        Console.Clear()
        For i = 0 To 4
            Console.Write(days(i))
            Dim a As Integer
            For a = 0 To 7
                If a <> 7 Then
                    Console.Write("    " & timetable(i, a))
                Else
                    Console.WriteLine("    " & timetable(i, a))
                End If
            Next
        Next
        ReadKey()
    End Sub
End Module

'T8Q6) Using a two-dimensional array, write a program that stores the names of ten
countries in column 1 And their capitals in column 2. The program should then pick a
random country And ask the user for the capital. Display an appropriate message to the
user to show whether they are right Or wrong. Expand the program above to ask the user 5
T8Questions And give a score of how many they got right out of 5.
```

```
Module T8Q6
    Sub Main()
        Dim array(,) As String = New String(,) {{"Pakistan", "Islamabad"}, {"India", "New
Delhi"}, {"USA", "Washington D.C"}, {"China", "Beijing"}, {"UK", "London"}, {"Russia",
"Moscow"}, {"France", "Paris"}, {"Netherlands", "Amsterdam"}, {"Bangladesh", "Dhaka"},
{"Saudi Arabia", "Riyadh"}}
        Dim correctAns As Integer = 0
        Dim rand As Random = New Random
        Dim r As Integer
        Dim i As Integer
        For i = 0 To 4
            r = rand.Next(0, 9)
            Console.WriteLine("What is the name of the capital city of " & array(r, 0))
            If Console.ReadLine() = array(r, 1) Then
                correctAns = correctAns + 1
            End If
        Next
        Console.WriteLine("Your total score is " & correctAns)
        ReadKey()
    End Sub
End Module

'T8Q7) Store in a 1-D array a set of 5 place names And in a 2-D array the distances
between the places. Ensure that the order of the places Is the same in both arrays. When
the names of 2 places are input, the distance between them Is displayed. If they are Not
both in the table, a suitable message should be displayed.
Module T8Q7
    Sub Main()
        Dim name() As String = New String() {"A", "B", "C", "D", "E"}
        Dim dist(,) As Double = New Double(,) {{0, 12, 23, 34, 21}, {12, 0, 21, 13, 7},
{23, 21, 0, 16, 27}, {34, 13, 16, 0, 40}, {21, 7, 27, 40, 0}} 'each array inside the main
array has the dist corresponding to the locations
        Dim place1, place2 As String
        Dim pos1, pos2 As Integer
        Console.WriteLine("Enter name of place 1")
        place1 = Console.ReadLine()
        Console.WriteLine("Enter name of place 2")
        place2 = Console.ReadLine()
        Dim i As Integer
        For i = 0 To 4
            If name(i) = place1 Then
                pos1 = i
            End If
            If name(i) = place2 Then
                pos2 = i
            End If
        Next
        Console.WriteLine("Distacne b/w " & name(pos1) & " and " & name(pos2) & " is " &
dist(pos1, pos2) & " km") 'pos1 gets the array(of the pos from which the dist is to be
calc) inside the main array and pos2 gets the element which represents the dist of that
place to place 2
        ReadKey()
    End Sub
End Module

'T8Q8) A Latin ST8Quare of order n Is an n x n array which contains the numbers 1, 2, 3,
..., n such that each row And column contain each number exactly once. For example the
following diagram shows a Latin ST8Quare of order 4. You can see that each row can be
```

```
obtained from the previous one by shifting the elements one place to the left. Design And
write a program to store such a Latin ST8Quare of a size given by the user. The program
should also display the Latin ST8Quare.

'Method1:
'Length(L) = 4

'L-3   L-2    L-1    L
'L-2   L-1    L      1
'L-1   L      1      2
'L   1 2      3

'Method2:
'Length = 4
'P = Previous

'1   2 3      4
'2   P+1      P+1    1
'3   P+1      1      P+1
'4   1 P+1    P+1
Module T8Q8Alt
    Sub Main()
        Console.WriteLine("Enter the length of latin sT8Quare")
        Dim maxnum As Integer = Console.ReadLine()
        Dim indexlen = maxnum - 1
        Dim LatinST8Quare(indexlen, indexlen)
        Dim numbering As Integer = 1
        For i = 0 To indexlen
            LatinST8Quare(0, i) = numbering
            LatinST8Quare(i, 0) = numbering
            numbering = numbering + 1
        Next
        For r = 1 To indexlen 'Starting from the 2nd row bcz the 1st row is already
filled
            For c = 1 To indexlen 'Starting from the 2nd column bcz the 1st column is
already filled
                If LatinST8Quare(r, c - 1) = maxnum Then
                    LatinST8Quare(r, c) = 1
                Else
                    LatinST8Quare(r, c) = LatinST8Quare(r, c - 1) + 1
                End If
            Next
        Next
        For r = 0 To indexlen
            For c = 0 To indexlen
                If c <> indexlen Then
                    Console.Write(LatinST8Quare(r, c) & " ")
                Else
                    Console.WriteLine(LatinST8Quare(r, c))
                End If
            Next
        Next
        ReadKey()
    End Sub
End Module
```

TOPIC 11:

```
'Pg62:
'Q1) Write a program that reads in a string And displays the number of characters in the
string.
Module T11Q1
    Sub Main()
        Dim chars() As Char = ReadLine().ToCharArray()
        ' ^"ToCharArray" Converts string into array with chars as elements
        WriteLine("Number of characters  = " & chars.Length)
        ReadKey()
    End Sub
End Module

'Q2) Write a program that displays the ASCII code for any given character.
Module T11Q2
    Sub Main()
        WriteLine("Enter a character")
        WriteLine(Asc(ReadLine()))
        ' ^"Asc()" Converts char in bracket to ASCII code
        ReadKey()
    End Sub
End Module

'Q3) Write a program that will display the character for a given ASCII code.
Module T11Q3
    Sub Main()
        WriteLine("Enter ASCII code")
        WriteLine(Chr(ReadLine()))
        ReadKey()
    End Sub
End Module

'Q4) Write a program that asks the user to type in a number with decimal places. The
program should then display the rounded And the truncated number.
Module T11Q4
    Sub Main()
        WriteLine("Enter a decimal number")
        Dim input As Double = ReadLine()
        WriteLine("Rounded number = " & Math.Round(input, 1,
MidpointRounding.AwayFromZero))
        WriteLine("Truncated number = " & Math.Truncate(input))
        ReadKey()
    End Sub
End Module

'Q5) Write a program that asks the user for their surname And displays the surname in
uppercase letters.
Module T11Q5
    Sub Main()
        WriteLine("Enter your surname")
        WriteLine(ReadLine().ToUpper())
        ReadKey()
    End Sub
End Module
```

```
'Q6) Write a program that displays today's date and formats the output appropriately.
Module T11Q6
    Sub Main()
        Dim Today As Date
        Today = Now()
        WriteLine("Date: " & Format(Today, "dd/mm/yy"))
        WriteLine("time: " & Format(Today, "hh:mm:ss"))
        ReadKey()
    End Sub
End Module

'Q7)Write a program that reads in a date, converts it into date format, adds a day And
displays the next day's date.
Module T11Q7
    Sub Main()
        Dim newdate As Date = Convert.ToDateTime(ReadLine())
        newdate = newdate.AddDays(1)
        WriteLine(newdate)
        ReadKey()
    End Sub
End Module

'Q8) Write a program that takes two letters as input And displays all the letters of the
alphabet between the two supplied letters (inclusive). For example, EJ produces EFGHIJ.
The letters are to be printed in the order in which the specified letters are supplied.
GB should produce GFEDCB. (TIP: Use the letters ASCII code – see page 53)
Module T11Q8
    Sub Main()
        Dim l1, l2 As Integer
        ' Asc() Converts letter to ASCII code
        WriteLine("Enter letter number 1")
        l1 = Asc(ReadLine)
        WriteLine("Enter letter number 2")
        l2 = Asc(ReadLine)
        If l1 <= l2 Then
            ' ^If letter1 less than or equal to letter2 then
            ' Looped in forward direction
            For i = l1 To l2
                WriteLine(Chr(i))
            Next
        Else
            ' ^If letter1 greater than letter2 then
            ' Looped in reverse direction
            For i = l1 To l2 Step -1
                WriteLine(Chr(i))
            Next
        End If
        ReadKey()
    End Sub
End Module

'Q9) Write a program that asks the user for a word, And then displays the ASCII code for
each letter in the word.
Module T11Q9
    Sub Main()
        WriteLine("Enter a word")
        Dim chr() As Char = ReadLine().ToCharArray()
        For i = 0 To chr.Length - 1
```

```
            ' ^"chr.Length - 1" becuz indexing starts from 0
            WriteLine(Asc(chr(i)))
            ' ^Converts the letter in the array to ASCII code
        Next
        ReadKey()
    End Sub
End Module

'Q10) Write a program that asks the user to enter a sentence, terminated by a full stop
And the pressing of the Enter key. The program should count the number of words And
display the result.
Module T11Q10
    Sub Main()
        WriteLine("Enter a sentence")
        ' Trim() gets rid of spaces at the start and end
        ' So that there is no prob in the code
        Dim input As String = Trim(ReadLine())
        WriteLine("Number of words in sentence = " & Split(input, " ").Length)
        ' ^"Split(a,b)" splits the array "a" with respect to the char b and forms an
array
        ' ".Length" gets the number of elements in the array.
        ' Hence the number of words are determined
        ReadKey()
    End Sub
End Module

'Q11) Write a palindrome tester. A palindrome Is a word Or sentence that reads the same
backwards as forwards. The user should enter a string And the program should display
whether the string Is a palindrome Or Not.
Module T11Q11
    Sub Main()
        WriteLine("Enter a word")
        Dim chr() As Char = ReadLine().ToCharArray
        ' ^"ToCharArray" Converts string into array with chars as elements
        Dim palindrome As Boolean = True
        For i = 0 To chr.Length - 1
            If chr(i) <> chr(chr.Length - 1 - i) Then
                ' ^i.e: 1st char from left side <> 1st char from right side
                ' If so then
                palindrome = False
            End If
        Next
        If palindrome = True Then ' If word is palindrome
            WriteLine("This word is a plaindrome word")
        Else ' If word is not palidrome
            WriteLine("This word is not a palindrome word")
        End If
        ReadKey()
    End Sub
End Module
```

TOPIC 13:

```
'Pg69:
```

```
'Q1) Write And test a procedure Swap, which takes two integers as parameters And returns
the first value passed to the procedure as the second value returned by the procedure And
vice versa.
Module T13Q1
    Sub Main()
        WriteLine("Enter 1st Integer")
        Dim a As Integer = ReadLine()
        WriteLine("Enter 2nd Integer")
        Dim b As Integer = ReadLine()
        ' "Swap()" is not a predefined function
        Dim array() As Integer = Swap(a, b)
        ' ^Declaration and assignment of array
        WriteLine("1st integer = " & array(0))
        WriteLine("2nd integer = " & array(1))
        ReadKey()
    End Sub

    Function Swap(ByVal a As Integer, ByVal b As Integer) As Integer()
        Return {b, a}
    End Function
End Module

'Q2) Write And test a procedure OutputSymbols, which takes two parameters: an Integer n
And a character symbol. The procedure Is To display, on the same line, the symbol n
times. For example, the Call OutputSymbols (5, ' # ' ) should display #####.
Module T13Q2
    Sub Main()
        WriteLine("Enter symbol")
        Dim s As String = ReadLine()
        WriteLine("Enter number of times you wish to output it")
        Dim n As Integer = ReadLine()
        OutputSymbols(n, s)
        ' ^Not a predefined function
        ReadKey()
    End Sub

    Sub OutputSymbols(ByVal n As Integer, ByVal s As String)
        For i = 1 To n ' Loop number of given times
            Write(s) ' Write on the same line
            ' No need to include WriteLine() the last time
            ' this loop runs becuz there is no need to write
            ' on the next line
        Next
    End Sub
End Module

'Q3) Write And test a procedure Sort, which takes two integers as parameters And returns
them in ascending order.
Module T13Q3
    Sub Main()
        WriteLine("Enter 1st Integer")
        Dim a As Integer = ReadLine()
        WriteLine("Enter 2nd Integer")
        Dim b As Integer = ReadLine()
        ' "Sort()" is not a predefined function
        Dim array() As Integer = Sort(a, b)
        ' ^Declaration and assignment of array
        WriteLine("1st integer = " & array(0))
```

```
        WriteLine("2nd integer = " & array(1))
        ReadKey()
    End Sub

    Function Sort(ByVal a As Integer, ByVal b As Integer) As Integer()
        If a > b Then
            Return {b, a}
        Else
            Return {a, b}
        End If
    End Function
End Module

'Q4) Write a program to let the computer guess a number the user has thought of, within a
range specified by you as the programmer.
Module T13Q4
    Sub Main()
        WriteLine("Think a number between 1 to 10 and press enter")
        ReadKey()
        Dim r As Random = New Random()
        WriteLine("You thought of " & r.Next(1, 11))
        ReadKey()
    End Sub
End Module

'Q5) The game 'Last one loses' is played by two players and uses a pile of n counters.
Players take turns at removing 1, 2 or 3 counters from the pile. The game continues until
there are no counters left and the winner is the one who does not take the last counter.
Using procedures, write a program to allow the user to specify n in the range 10 – 50
inclusive and act as one player, playing at random until fewer than 5 counters remain.
Try playing against your program, and then playing to win.


'Q6) Create a procedure GetLotteryNumbers that will supply 6 unique random numbers
between 1 And 49. One possible strategy, Or algorithm, Is:
'Initialise an array by Using a For Loop To store the values 1 To 49
'Repeatedly select a random element from array until a non-zero value Is selected
'Display this value
'Set that element To zero
'Repeat the above three steps until six numbers have been selected.
Module T13Q6
    Sub Main()
        Dim list As List(Of Integer) = New List(Of Integer)
        Dim r As Random = New Random
        Dim i As Integer
        For i = 0 To 48
            list.Add(i + 1)
        Next
        For i = 0 To 5
            Dim val As Integer = r.Next(list(0), list(list.Count - 1))
            list.Remove(val)
            WriteLine(val)
        Next
        ReadKey()
    End Sub
End Module
```

TOPIC 14:

```
'Pg71:
'Q1) Write a function to convert temperatures from Fahrenheit to Celsius. The function should take one integer parameter (the temperature in Fahrenheit) And return a real result (the temperature in Celsius). The formula for conversion Is: Centigrade = (Fahrenheit – 32) * (5 / 9)
Module T14Q1

    Sub Main()
        WriteLine("Enter temprature in Fahrenheit")
        WriteLine(Conv(ReadLine()))
        ' ^Takes the input from user + Converts it + Displays it
        ReadKey()
    End Sub

    Function Conv(ByVal f As Double) As Double
        Return (f - 32) * (5 / 9)
    End Function
End Module

'Q2) Write a function that converts a string passed as a parameter into a capitalized string. See page 46 for details on string manipulation)
Module T14Q2

    Sub Main()
        WriteLine("Enter a string")
        WriteLine(Conv(ReadLine()))
        ' ^Takes the input from user + Converts it + Displays it
        ReadKey()
    End Sub

    Function Conv(ByVal s As String) As String
        Return s.ToUpper()
    End Function
End Module

'Q3) Write a function that returns the total number of seconds, calculated from a whole number of hours, minutes And seconds provided as 3 parameters.
Module T14Q3
    Sub Main()
        Dim h, m, s As Integer
        WriteLine("Enter number of hours")
        h = ReadLine()
        WriteLine("Enter number of minutes")
        m = ReadLine()
        WriteLine("Enter number of seconds")
        s = ReadLine()
        WriteLine("Number of hours = " & Conv(h, m, s))
        ReadKey()
    End Sub

    Function Conv(ByVal h As Integer, ByVal m As Integer, ByVal s As Integer) As Integer
```

```
        Return (h * 3600) + (m * 60) + s
    End Function
End Module

'Q4) Write your own random function RandomNumber that returns values in the range from 1
to the integer supplied as parameter. Tip: use the Maths library (see page 18)
Module T14Q4
    Sub Main()
        WriteLine("Enter range")
        WriteLine("Random Number = " & RandomNumber(ReadLine()))
        ' ^Takes input + Calcs Random Num + Displays it
        ReadKey()
    End Sub

    Function RandomNumber(ByVal x As Integer) As Double
        Dim r As Random = New Random()
        Return r.Next(1, x)
    End Function
End Module

'Q5) Write a tables tester. The program chooses 2 random numbers And asks the user what
Is the product of these 2 numbers. If the user gets the answer wrong, the correct answer
should be displayed. The program should ask 10 questions And then display a score out of
10 And a suitable message.
Module T14Q5
    Sub Main()
        Dim r As Random = New Random
        Dim n1, n2, score As Integer
        score = 0
        For i = 0 To 9
            ' Get 2 random nums
            n1 = r.Next(1, 11)
            n2 = r.Next(1, 11)
            WriteLine("Enter product of " & n1 & " and " & n2)
            If ReadLine() = n1 * n2 Then
                ' ^Gets input + Compares it with the product
                ' If correct then score is increased
                WriteLine("Correct Ans")
                score = score + 1
            Else
                ' If wrong then score is decreased
                WriteLine("Incorrect Ans. Correct Ans is " & n1 * n2)
            End If
        Next
        WriteLine("Your Score = " & score)
        ReadKey()
    End Sub
End Module
```